MODULE 14 · DOMAIN 2 - RELIABILITY AND BUSINESS CONTINUITY

Multi-AZ, Failover and Health Checks

RDS, Aurora, EFS and Route 53 - what survives an Availability Zone failure, and how long it takes.

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Multi-AZ answers one question: **does the service survive losing a whole Availability Zone?** Everything else is how fast, and what breaks meanwhile.

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Multi-AZ DB instance or Multi-AZ DB cluster?

MULTI-AZ DB INSTANCE

One hidden standby

- One synchronous standby in another AZ
- The standby serves NO traffic at all
- Failover typically 60 to 120 seconds
- All engines, including Oracle and SQL Server

MULTI-AZ DB CLUSTER

Two readable standbys

- One writer plus two readable standbys
- Spread across three Availability Zones
- Failover typically under 35 seconds
- MySQL and PostgreSQL only

THE TIEBREAKER

The classic Multi-AZ standby is invisible. You cannot read from it, connect to it, or take a backup from it in the way people expect.

Multi-AZ or a read replica?

MULTI-AZ IS FOR

Availability

- Synchronous replication, no data loss on failover
- Automatic failover, same endpoint
- Standby in a different AZ, same Region
- Does not improve read performance at all

READ REPLICAS ARE FOR

Scale

- Asynchronous replication, so replica lag exists
- Manual promotion, and a different endpoint
- Same AZ, another AZ, or another Region
- Offloads read traffic from the primary

THE TIEBREAKER

This is the single most confused pair in the RDS syllabus. Availability versus scale, synchronous versus asynchronous, automatic versus manual.

What actually happens during an RDS failover

- The **DNS endpoint is repointed** at the standby. Your connection string never changes.
- **Existing connections are dropped.** The application must reconnect, so connection pools and retry logic matter.
- Triggers: AZ outage, primary instance failure, storage failure, instance type change, or OS patching.
- **Reboot with failover** forces one deliberately. That is how you test, and how you apply some changes with one controlled outage.

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>
> Applications that cache the resolved IP address rather than honouring DNS TTL will keep failing after the database has recovered. That is an application bug, not an AWS one.

Aurora is built differently

- Storage is a **shared cluster volume**, six copies across three AZs. Compute instances attach to it rather than owning a copy.
- Up to **15 replicas**, all reading the same volume, so replica lag is typically well under a second.
- **Failover promotes a replica**, typically in under 30 seconds. With no replica, Aurora creates a new instance, which takes far longer.
- **Failover priority tiers 0 to 15** decide who is promoted. Lowest tier first; on a tie, the largest instance.

> ON THE EXAM
>
> An Aurora cluster with no replica is not highly available. The storage survives, but there is nothing to promote, so recovery means launching a new instance.

Aurora endpoints

- **Cluster endpoint** always points at the current writer. Use it for writes, and it follows failover automatically.
- **Reader endpoint** load-balances across all available replicas. Use it for read traffic.
- **Custom endpoints** group a chosen subset, for example sending reporting queries only to two large replicas.
- **Instance endpoints** address one specific instance. Useful for diagnosis, dangerous in application configuration.

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> An application pointed at an instance endpoint will not follow a failover. That is the Aurora equivalent of hard-coding an IP address.

Failover times, side by side

DEPLOYMENT	TYPICAL FAILOVER
RDS Multi-AZ DB instance	60 to 120 seconds
RDS Multi-AZ DB cluster	Under 35 seconds
Aurora, with at least one replica	Under 30 seconds
Aurora, no replica	Minutes. A new instance must be created
Read replica promotion	Manual, and it is not a failover mechanism

ON THE EXAM

Replica lag bounds failover time. A promoted instance must apply outstanding transactions before it can accept writes.

EFS and the rest of the stack

- **EFS Regional** stores data across multiple AZs and survives losing one. **EFS One Zone** does not, and is cheaper.
- Clients mount through an **AZ-specific mount target**. Create one in every subnet your instances use.
- A **NAT gateway lives in one AZ**. One NAT gateway means an AZ failure removes outbound internet for every private subnet routed to it.
- **ElastiCache** needs Multi-AZ with automatic failover enabled, and replicas in other zones, to survive a zone loss.

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> The database being Multi-AZ does not make the system Multi-AZ. A single NAT gateway or a single-AZ cache defeats the whole design quietly.

Static stability

- A statically stable system **keeps working during a failure without needing to change anything**.
- Pre-provision capacity in every AZ to handle the loss of one, rather than relying on scaling out during the incident.
- Recovery that depends on the control plane is fragile, because control planes are often the busiest thing during a large event.
- Three AZs at 50 percent utilisation survives losing one at 75 percent. Two AZs at 50 percent becomes 100 percent, with no headroom.

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> "It will just scale out when the AZ fails" assumes launching instances works during the exact moment everyone else is also launching instances.

Route 53 health checks: three kinds

- **Endpoint health checks** probe an HTTP, HTTPS or TCP endpoint from checkers around the world.
- **CloudWatch alarm health checks** take their state from an alarm. This is how you health-check something private.
- **Calculated health checks** combine other health checks with boolean logic: healthy if at least 2 of 3 are healthy.
- **Endpoint checks run from the public internet**, so they cannot reach anything inside a VPC. That is not a configuration problem.

> ON THE EXAM
>
> "Health check a private resource" always means the CloudWatch alarm type. There is no way to make a public checker reach a private subnet.

How fast DNS failover really is

- **Request interval**: 30 seconds standard, or 10 seconds fast. **Failure threshold**: 1 to 10, default 3.
- Detection time is roughly **threshold multiplied by interval**. Then add the record's **TTL**.
- Then add **resolver caching** and **client-side connection pinning**, which you do not control.
- Standard 30-second checks with threshold 3 and a 60-second TTL can mean **two to four minutes** of user-visible failure.

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> "Our TTL is 60 seconds so failover takes a minute" is true at the authoritative layer and misleading as a promise to users.

Health check details that surprise people

- When you specify an endpoint **by domain name**, Route 53 health checks use **IPv4 only**. An IPv6-only domain fails DNS resolution.
- An HTTPS health check confirms a response, not certificate hygiene. Pair it with a **separate ACM certificate expiry alarm**.
- **String matching** checks the first 5,120 bytes of the body, so a health page can assert on real content rather than just a 200.
- For weighted, latency and multivalue answers, **if every record is unhealthy Route 53 responds as though all were healthy**.

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> That last behaviour is deliberate. Returning nothing helps nobody, so Route 53 would rather hand back a possibly-broken answer than no answer at all.

Evaluate target health, or a separate health check?

EVALUATE TARGET HEALTH

For alias records

- Alias to an ALB, NLB, S3 website or another record
- Uses the target's own health directly
- No health check to create, and no monthly charge
- Simplest correct answer for an ALB alias

A SEPARATE HEALTH CHECK

When you need more

- Non-alias records, or an IP address target
- Custom path, string matching, or a fast interval
- Calculated logic across several signals
- CloudWatch alarm type, for private resources

THE TIEBREAKER

For an alias to a load balancer, evaluate target health is usually right and often forgotten. It inherits health that already exists.

Three different things called a health check

- **Auto Scaling health check** decides whether to **replace an instance**. EC2 status checks, ELB health, or custom.
- **Target group health check** decides whether to **send a request to a target**. Removes it from rotation; does not terminate it.
- **Route 53 health check** decides **which DNS answer to return**. Operates at the record level, not the instance level.
- They chain: an ELB health check can feed the ASG's decision, and Route 53 can evaluate the load balancer's health.

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> Replace it, stop sending to it, or stop resolving to it. Three different verbs, and a scenario always tells you which one it wants.

Failover routing in practice

- **Failover routing** uses a primary and a secondary record. The secondary is returned when the primary's health check fails.
- **Active-passive**: the secondary is a standby site. **Active-active**: use weighted or latency routing with health checks on every record.
- A common secondary is a **static S3 or CloudFront maintenance page**, so users see something honest rather than a timeout.
- Test failover deliberately. A secondary that has never served traffic is a hypothesis, not a plan.

> ON THE EXAM
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> Failover routing needs exactly one primary and one secondary. For more than two targets, or for load sharing, use weighted or latency routing with health checks.

An RDS for PostgreSQL database is a Multi-AZ deployment. Reporting queries are slowing the application, and a developer proposes pointing the reporting tool at the standby instance to offload the reads.

The team also wants failover to complete faster than the current 90 seconds, and reporting must not affect the writer.

What should the team do?

A Point the reporting tool at the standby, as proposed

B Convert to a Multi-AZ DB cluster, and send reporting to the reader endpoint

C Create a read replica and leave the Multi-AZ deployment as it is

D Increase the instance class of the primary

◆ PAUSE AND DECIDE

What should the team do?

A Read from the standby

Impossible. The classic Multi-AZ standby serves no traffic at all and has no endpoint to connect to.

B Multi-AZ DB cluster

Two readable standbys give the read offload, and failover drops from 60-120 seconds to under 35. Both requirements, one change.

C Read replica alongside Multi-AZ

The strongest alternative: it does solve the reporting offload. It does nothing about the 90-second failover requirement.

D Bigger primary instance

Buys headroom while leaving reporting on the writer. It also does not change failover time at all.

THE SIGNAL

Two requirements again. Reporting offload alone would allow option C; the failover requirement is what forces the cluster.

A team configures Route 53 failover routing between two Regions. The primary record has a health check against an internal application load balancer in a private subnet. The health check has been permanently unhealthy since it was created, so all traffic goes to the secondary Region.

The load balancer and its targets are confirmed healthy.

What is the cause?

A The failure threshold is set too low

B Route 53 endpoint health checkers run on the public internet and cannot reach a private endpoint

C The record's TTL is too high

D Failover routing requires a calculated health check

◆ PAUSE AND DECIDE

What is the cause?

A Failure threshold too low

Would cause flapping on intermittent problems, not a check that has never once succeeded since creation.

B Public checkers cannot reach a private endpoint

Endpoint health checks originate from the public internet. An internal load balancer has no public address, so every probe fails.

C TTL too high

TTL affects how long clients cache an answer. It has no bearing on whether a health check succeeds.

D Calculated check required

Invented requirement. Failover routing works with any health check type, including none at all.

THE SIGNAL

Never healthy since creation is the signal. Intermittent problems flap; a structural impossibility fails from the first probe.

An architecture runs across three Availability Zones: EC2 instances in private subnets in each, an Auto Scaling group spanning all three, an ALB across all three, and an RDS Multi-AZ database. There is a single NAT gateway in the first AZ.

During a simulated failure of the first AZ, instances in the other two zones lose the ability to reach an external payment API, and the application fails.

What should be changed?

A Deploy a NAT gateway in each Availability Zone, with each private subnet's route table pointing to its local one

B Add a second ALB in another Availability Zone

C Convert the RDS deployment to a Multi-AZ DB cluster

D Enable cross-zone load balancing on the ALB

◆ PAUSE AND DECIDE

What should be changed?

A A NAT gateway per AZ

A NAT gateway lives in one AZ. Routing all private subnets through one makes that zone a single point of failure for all outbound traffic.

B A second ALB

An ALB already spans the zones you enable. The load balancer was never the failing component here.

C Multi-AZ DB cluster

Improves database failover time. The database was already Multi-AZ and is not what broke.

D Cross-zone load balancing

Always on for an ALB anyway, and it concerns inbound distribution rather than outbound internet access.

THE SIGNAL

The scenario carefully lists everything that IS multi-AZ, then names one thing that is not. That asymmetry is the whole question.

> A team needs user-visible failover to a secondary Region within 60 seconds. They use Route 53 failover routing with a standard 30-second health check interval. The failure threshold is 3, and the record TTL is 300 seconds.
>
> Testing shows failover taking between four and six minutes.

Which combination brings this closest to the target? (Choose two.)

A. Change the health check to a 10-second fast interval and reduce the failure threshold to 2

B. Reduce the record TTL from 300 seconds to 60 seconds or lower

C. Switch from failover routing to weighted routing

D. Enable evaluate target health on the primary record

◆ PAUSE AND DECIDE

Which combination brings this closest to the target?

A Fast interval, threshold 2

Detection drops from 90 seconds to 20. The single largest saving available on the detection side.

B Reduce the TTL

300 seconds of caching dominates everything else. No detection improvement matters while clients cache the old answer for five minutes.

C Weighted routing

Changes how traffic is distributed, not how quickly a failure is noticed or how long answers are cached.

D Evaluate target health

Removes the need for a separate health check on an alias. It does not make detection faster, and here a custom check is wanted.

THE SIGNAL

Failover time is detection plus caching. Fixing only one leaves the other dominating, which is why this is a "choose two".

An Aurora PostgreSQL cluster runs a single writer instance with no replicas. The team believes it is highly available because Aurora stores six copies of the data across three Availability Zones.

During an AZ failure the database was unavailable for several minutes, and the team wants to know why.

What explains this, and what is the fix?

A Aurora storage is not multi-AZ; enable Multi-AZ on the cluster

B With no replica there is nothing to promote, so Aurora had to create a new instance. Add at least one replica in another AZ

C The cluster endpoint had cached the failed instance's IP address; reduce the TTL

D Aurora Serverless v2 is required for automatic failover

◆ PAUSE AND DECIDE

What explains this, and what is the fix?

A Storage is not multi-AZ

The team's premise is correct: the storage genuinely is spread across three AZs. Durability was never the problem.

B No replica to promote

Storage survived; compute did not. Fast failover means promoting an existing replica, and there was none to promote.

C Endpoint cached an IP

The cluster endpoint follows failover. The delay was creating a writer, not resolving to one.

D Serverless v2 required

Invented requirement. Provisioned Aurora fails over automatically, provided there is a replica available.

THE SIGNAL

Durable is not the same as available. The data was safe throughout; there was simply no compute able to serve it.

The five things to carry forward

- **Multi-AZ is availability, replicas are scale.** The classic standby is invisible and serves nothing.
- **60-120 seconds for a Multi-AZ instance, under 35 for a Multi-AZ cluster, under 30 for Aurora with a replica.**
- **Aurora with no replica is not highly available.** Durable storage is not the same as available compute.
- **Route 53 endpoint checks cannot reach private resources.** Use the CloudWatch alarm type.
- **Replace it, stop sending to it, stop resolving to it.** Auto Scaling, target group, and Route 53 health checks.

> ON THE EXAM
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> Next: scaling the data tier - read replicas, Aurora Serverless v2, DynamoDB modes and caching.